

4-Bit Password Encryption & Verification

Jubayer Iqbal, 24101512
Mehedi Hasan Somoy, 24101093
Yeasin Arafat Rabby, 24101120
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh

I. ABSTRACT

This project presents a 4-bit password encryption and verification system implemented using digital logic components. The system uses XNOR gates to encrypt a user-defined 4-bit password with a 4-bit encryption key, while D-type flip-flops are used to store the encrypted password. While verifying, the entered password is encrypted using the same key and compared with the stored encrypted data. The system provides a visual indication of the verification result using green and red LEDs for correct and incorrect passwords, respectively. The circuit was implemented on a breadboard and tested with different 4-bit password combinations to verify its functionality.

II. IMPLEMENTATION AND WORKING PRINCIPLE

The proposed 4-bit password encryption and verification system is implemented using digital logic components to provide a simple and reliable method for password storage and verification. The system operates in two stages: password storage and password verification.

During the password storage stage, the user first enters a 4-bit password using the input switches and sets a 4-bit encryption key using the encryption-key switches. When the store button is pressed, the four password bits are processed by the encryption logic together with the corresponding encryption-key bits. The encrypted bits are then loaded into the four D-type flip-flops [1], where each flip-flop stores one bit of the encrypted password.

The IC 7474 Dual D-Type Flip-Flop is used to hold the encrypted password [1]. Each IC contains two D-type flip-flops. Each flip-flop is able to keep one bit of information. When a clock pulse is given, the bit at the D input is moved to the Q output and maintained until the next clock signal is given. In this project, four flip-flops are used to hold the four bits of the encrypted password.

TABLE I
TRUTH TABLE FOR XNOR GATE

A	B	A XNOR B
0	0	1
0	1	0
1	0	0
1	1	1

The IC 4077 Quad 2-Input XNOR Gate is used in the encryption and verification circuitry [2]. Each XNOR gate has two inputs and produces a HIGH output when the two input bits are same, and a LOW output when they are different (TABLE I). In the verification stage, the XNOR gates compare each bit of the newly encrypted password with the corresponding stored bit. The output is later processed by logic circuitry to determine whether the entered password matches the stored password.

For each bit, the comparison circuit produces a HIGH output when the two encrypted bits are same and a LOW output when they are different. The individual comparison outputs are then combined to produce the final verification result. If all four encrypted bits match the stored encrypted password, the final output becomes HIGH, causing the green LED to turn on and indicating that the entered password is correct. If any of the four bit differs, the final output becomes LOW, causing the red LED to turn on and indicating that the entered password is incorrect.

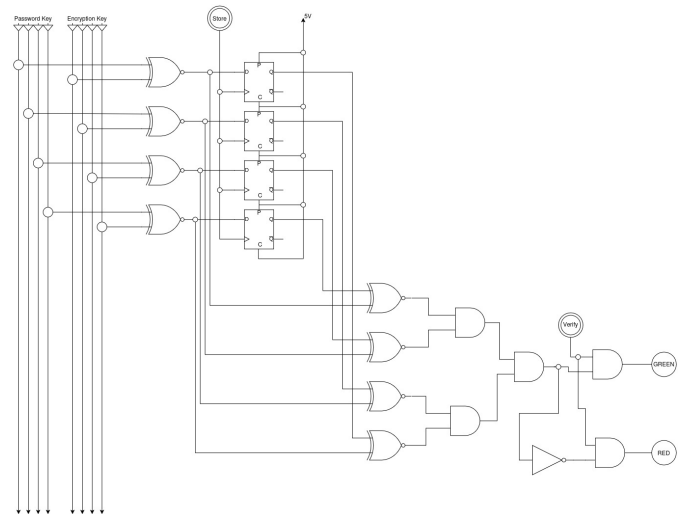


Fig. 1. Circuit diagram of the proposed system.

The complete circuit diagram of the proposed 4-bit password encryption and verification system is shown in Fig. 1. It illustrates the interconnection of the input, encryption, storage, verification, and output stages.

TABLE II
RELEVANT INTEGRATED CIRCUITS

Component	Purpose
IC 7474	Password Storage
IC 4077	Password Comparison
IC 7408	Logic Circuitry
IC 7404	Logic Circuitry

One of the challenges we faced during the hardware implementation was dealing with floating inputs. When a switch was not providing a defined HIGH or LOW logic level, the corresponding input could produce unpredictable outputs. This issue was resolved by using appropriate pull-up or pull-down resistors to ensure that each input remained at a defined logic state when the switch was not activated.

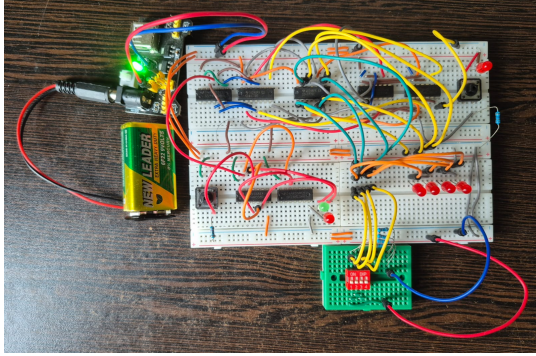


Fig. 2. Completed hardware implementation of the proposed system

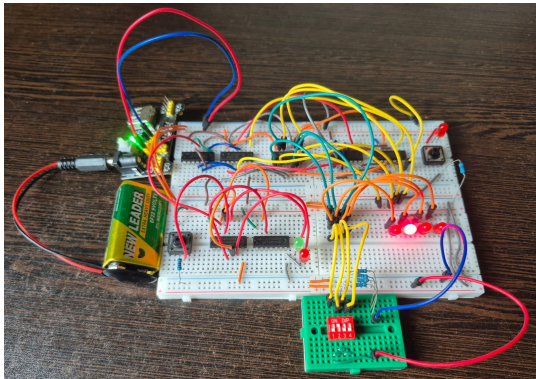


Fig. 3. Completed hardware implementation of the proposed system

The completed hardware implementation of the proposed system is shown in Fig. 2 and Fig. 3. The photographs illustrate the breadboard arrangement and the connections between the input, encryption, storage, verification, and output sections.

III. CONCLUSION

A 4-bit password encryption and verification system was successfully implemented using digital logic components. XNOR gates were used for both password encryption and bit-by-bit comparison. To store the encrypted password D-type flip-flops were used. The verification circuit successfully

determines whether the newly entered password matches the stored password and provides a clear visual indication through green and red LEDs.

As future improvements, a more complex encryption key could be implemented to enhance the security of the system. A lockout mechanism could also be added to prevent repeated unauthorized attempts. Additionally, a display could be incorporated to provide clearer feedback to the user regarding the verification status.

REFERENCES

- [1] Futurlec, "7474 Dual D-Type Flip-Flop," *Futurlec*, accessed Aug. 31, 2026. [Online]. Available: <https://www.futurlec.com/74/IC7474.shtml>
- [2] Build Electronic Circuits, "IC 4077: Quad 2-input XNOR gate," *Build Electronic Circuits*, accessed Aug. 31, 2026. [Online]. Available: <https://www.build-electronic-circuits.com/4000-series-integrated-circuits/ic-4077/>